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T4 _____	**C**	F4 _____

2016

MCM/ICM

Summary Sheet

(Your team's summary should be included as the first page of your electronic submission.)

Type a summary of your results on this page. Do not include the name of your school, advisor, or team members on this page.

We develop a model to determine an optimal investment strategy to improve the performance of undergraduate students in the US. Our model has three parts:

In the first part, we collect data about the focus of other foundations' investment by subjects and locations. We consider the charitable identity of the Goodgrant as well. Then we set out to decide our focus, which is to invest more on those schools with more minority races, lower educational performance, higher debt ratio and so on. In this part, we also classify the data into two groups, one for school selecting, and another for ROI determining.

In the second part, as a data extraction, we build a efficient and intuitive model to rank the candidate schools in accordance with the correlation of our focus, using the PCA method. After that, the top 50 schools are selected as our target schools.

In the third part，we make a key assumption that the social utility of a school has logarithmic relationship with the earnings of graduated students and the graduation rate. More over, we create a parameter k to denote the marginal rate of substitution (MRS) between the two factors above. After that, we come to define the ROI function of each target school as the incremental utility.

We further discuss how to devise the best strategy with several methods. At last, we choose the improved PSO algorithm based on augmented Lagrange function. This algorithm is a typical method to solve the multivariable optimization problem with constraint conditions. Then we offer a recommending list by the cumulative ROI in five years. What's more, our model is broad enough to accommodate any non-linear constraint optimization problem.

Finally, we change the numerical value of parameter k to examine the sensitivity of our investment strategy. The result shows that our model is robust.

The Optimal Investment Strategy Based on the Large-scale Non-linear Constraint Optimization Methods

Contents

1 Problem Statement 3
2 Planned Approach 3
3 Assumptions 4
4 Data Analysis and Focus Decision 5
4.1 Data Analysis 5
4.2 Focus Decision 6
5 School Selecting 8
5.1 Manual Selection 8
5.2 PCA Selection 8
5.2.1 Standardization 8
5.2.2 Calculation 9
5.2.3 Principle Components 9
5.2.4 PCA Results 9
6 Strategy Making 10
6.1 The ROI Function 11
6.2 Optimizing the Total ROI 12
6.2.1 Karush-Kuhn-Tucker Conditions 14
6.2.2 PSO Algorithm 16
6.2.3 Improved PSO algorithm based on augmented Lagrange function (LA_PSO_GT) 16
7 Result 16
7.1 Optimal Investment Strategy and Recommending List 17
8 Testing our Model 18
8.1 Sensitivity Analysis 18
8.2 Strengths 19
8.3 Weaknesses 19
9 Conclusion 20
10 Letter to the CFO of the Goodgrant Foundation 20
11 References 22
12 Appendix 1: Recommending List 23
13 Appendix 2 An Introduction to the Improved PSO Algorithm Based on Augmented Lagrange Function 25

1 Problem Statement

Private foundations are created by an individual, family, or business to fulfill specific charitable missions. Those like Gates foundation and Lumina foundation make great efforts to improve the quality of health and education in relatively poor areas. We must set big goals and spare no effort on the way because the world won't get better by itself. The Goodgrant, one of the foundations, intends to help improving educational performance of undergraduates attending colleges and universities in the United States. Given its potential donation of 100 million dollars per year in five years, what is the best investment strategy? We are tasked with creating models that can be applied in the universities across the nation. The solution proposed within this paper will offer an insight to use the big data and will objectively devise the investment strategy including the target schools, investment amount and duration.

2 Planned Approach

Our objective is to set out the best strategy including three components:(1) target schools;(2) the investment amount per school; (3) the investment duration. And also we will offer an optimized and prioritized recommendation list of candidate schools based on each school's return on investment (ROI).

Faced with the big data problem, we can't use the data directly because of the limitation of our personal computers and the length of the contest. If the data are directly applied, the computing system will run several days or weeks. As a result, the data selection is extremely important, which will also reflect the focus of the foundation. To determine the most effective computing system, we divide the problem into three parts together with the procedures as follows:

Part one: Data Analysis and Focus Decision

1. We will give an analysis of the big data of the problem, which includes information of near 3000 schools.
2. Based on the data given and the statistics of the focus of foundations collected from the Internet, we will decide the focus of the Goodgrant, avoiding duplicating the investment and focus of other large grant organizations.

Part two: School Selecting

1. Manual selection. We have taken some schools out of consideration for certain reasons (the reason will be explained below). For example, we exclude the schools located at NY, CA, WA and MA due to the large amount of existing grant foundations.
2. PCA (principle component analysis) selection. According to part of the data, the PCA method can rank the candidate schools by the degree of correlation of our focus. The top 50 schools will be selected out.

Part three: Strategy Making

1. Derive a ROI function that, given the year and a specific investment amount of a candidate school, can output the utility in an appropriate manner. The function is based on the graduation rate, earnings of graduated students and so on.
2. Utilize an optimization algorithm to maximize the total utility of the target 50 schools (in part two (2)), return the amount of investment and the time duration per school.

3 Assumptions

Due to limited data about the educational performance of the candidate schools, the performance of the undergraduate students and the specific distribution of other grants by subjects, races and locations, we use the following assumptions to complete our model. These simplified assumptions will be used through our paper and can be improved with more reliable data.

- The statistics of the candidate schools can be regarded as constant within five years. This assumption is reasonable to a large extent because the identities of a specific college won't change a lot in five years.
- The school will devote all the funding received this year to improving the students' performance.
- The appropriate manner to measure the return on the investment is the school's incremental utility. The utility function must be concave ($\partial^2 y/\partial x^2$<0). If not, we should give the whole 100 million to one college to maximize the total incremental utility, which is opposite against the common sense. And it's reasonable in economic consideration, as with the capital growth, the marginal production will be less and less. So we assume the utility function has this typical formulation $U_i = \log(x)$, where U_i is the utility, x are the independent variables.
- Neglect the discount rate of the capital. In this paper, we do not take the

inflation into consideration.

Here are the notations and their meanings in our paper:

Notation	*Meaning*
U	*Social utility of the school*
x	*Amount of the investment*
j	*Time period (j =0,1,2,3,4,5)*
m	*Share of students earning over $25,000/year*
M	*Median earnings of students working and not enrolled 10 years after entry*
N	*Number of undergraduate degree-seeking students*
k	*MRS (marginal rate of substitution)*
g	*Graduation rate*
λ, ν	*Lagrange multiplier*

Table 1: notation

4 Data Analysis and Focus Decision

Since we are tackling with a problem with big data, there is a diversity of inputs with different types. On the other hand, the inputs interact with each other to some degree. We must deeply analyze the data to dig out the meaning of each column and separate them in different groups.

After the analysis, we set out to collect the data of other large grant foundations，including their focus by different subjects, races and locations, together with other information available. Based on the data collected, we can determine the focus of the Goodgrant, which ensures the least degree of duplication.

4.1 Data Analysis

We analyze the data in the attached Excel sheet *Most Recent Cohorts Data*. We find out that there are continuous and discrete data. The continuous data can be separated into two groups, one for determining the focus as well as school selecting, another for measuring the school's utility as well as determining the ROI. So at last, we separate them into three components, each for different use.

Type and use	*Variable*
Continuous data for school selecting	*SAT/ACT scores*
	PCIP (the distribution by subject)
	UGDS_(the distribution by race)
	PPTUG_EF(part-time ratio)

	PCTPELL(pell grant ratio)
	UG_25_abv(the percentage of students above 25)
	GRAD_DEBT_MDN_SUPP(debt)
	RPY_3YR_RT_SUPP(debt ratio)
Discrete data for school selecting	*CONTRAL PBI ANNHI TRIBAL AANAPII HIS NANTI*
Continuous data for determining the ROI	*UGDS (the number of students)*
	C150 C200 (graduation rate)
	md_earn_wne_p10(median of earnings)
	gt_25k_p6(earnings above 25k ratio)

Table 2: the different types of variables and their use

4.2 Focus Decision

First of all, we define the Goodgrant as a charitable organization to make the world more equal. Based on this duty, the Goodgrant aims at helping those undergraduate students who are relatively poor, under debt, or has low SAT/ACT scores and those schools located at small cities, consisting of more minority races, with low graduated rate and so on.

Secondly, for not duplicating other organizations' investment and focus, we collect the data of the distribution of 1000 foundations by location and subject. The statistics are presented below.

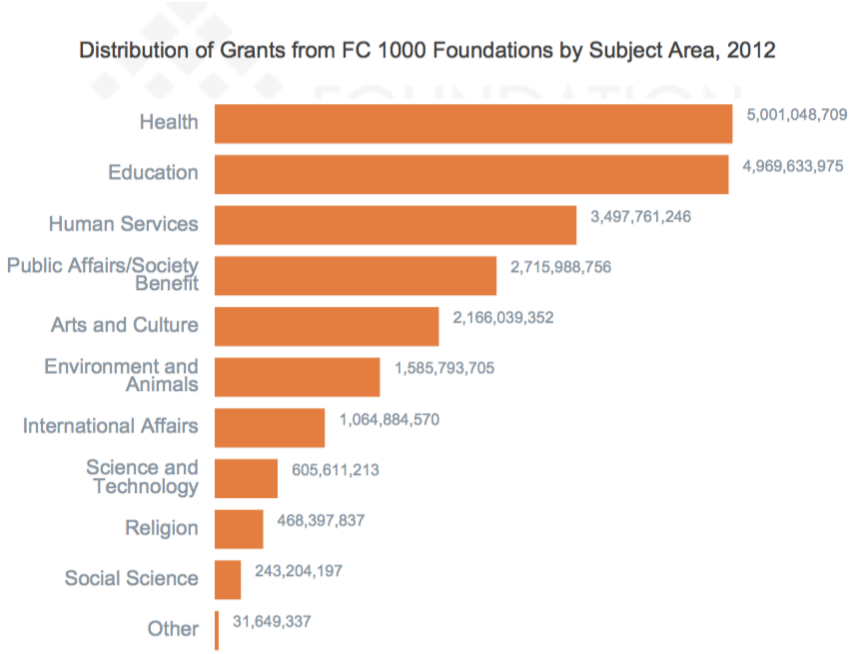


Figure 1: distribution of grants from FC 1000 foundations by subject area [1]

Source: http://data.foundationcenter.org/#/fc1000/subject:all/all/total/bar:amount/2012

We can see that most foundation invested in the following three subjects:

Health, Education and Human services from Figure 1. We define these three subjects as strong subjects. Notice that we have already the PCIP data (the percentage of degree awarded in different subjects). Compared to other subjects, the above three major subjects have received a large amount of grants. Further, the students who major in these three subjects will benefit a lot. So we decide to invest less on these three subjects but more on other subjects.

However, there is a problem behind the logic above. The large amount of capital invested on these three subjects doesn't mean that we should invest less on them. We only consider the capital supply while neglecting the capital demand. In other word, the financing gap really counts. However, taking capital demand into consideration will make the problem even complex. Also, we find it difficult to acquire the data of capital demand. As a result, we only consider the supply of capital, reluctantly.

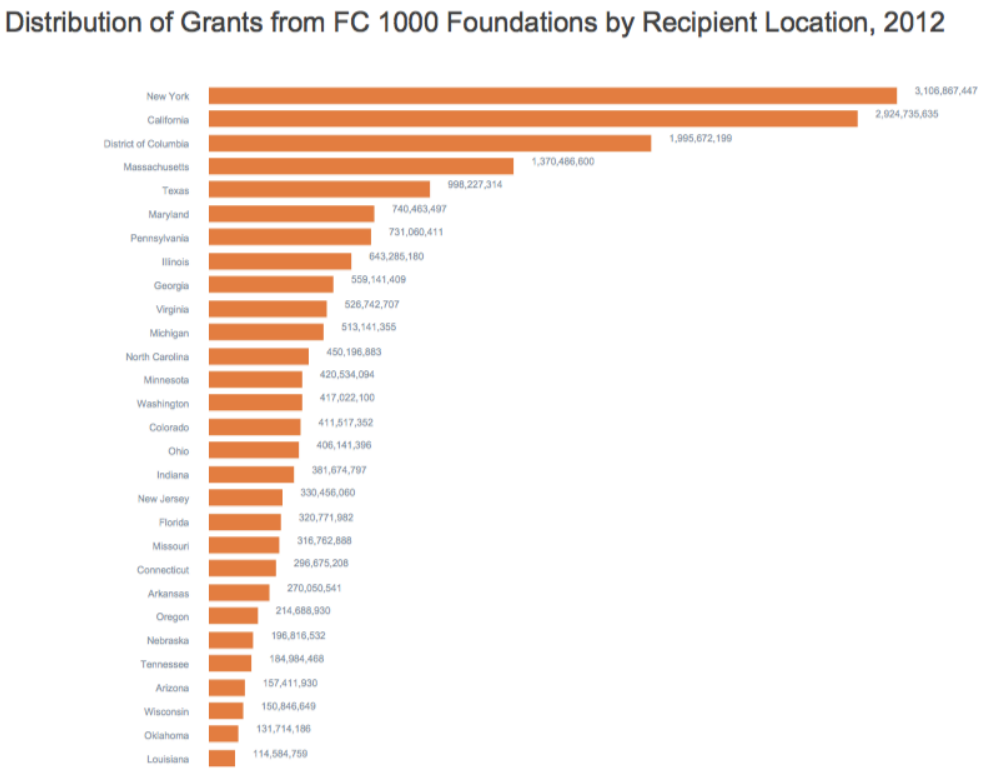


Figure 2: distribution of grants from FC 1000 foundations by recipient location

Source: http://data.foundationcenter.org/#/fc1000/subject:all/all/total/bar:amount/2012

Figure 2 tells us that most of the recipients are in four states: New York, California, District of Columbia and Massachusetts. Follow the focus of the Goodgrant, we decide not to invest on these four states because these four states are relatively wealthy and already have many foundations. However, the problem also exists which is similar to the above.

Here is a summary of our focus. We will invest more on schools which are/have:

1. Lower SAT/ACT scores
2. Lower percentage of students who receive pell grant
3. Higher percentage of minority races
4. Higher percentage of degree awarded in week subjects
5. Higher part-time ratio
6. Higher debt/loan

5 School Selecting

Based on the data analysis and our focus explained above, we intend to reduce the number of candidate schools from 2977 to 500 by manual selection and PCA selection. The smaller number of schools will make the algorithm easier to run in the next part "strategy making".

5.1 Manual Selection

According to the analysis and explanation above, we don't invest on the schools in four states: New York, California, District of Columbia and Massachusetts. So we directly delete the schools in these four states from our candidate sheet. This step reduces the number from 2977 to 2323.

5.2 PCA Selection

In this part, we intend to rank the schools based on the correlation between the school's identities and our focus. Since we make use of 68 indicators with complex interactions, we ought to extract a few principle indicators (usually 3 or 4) from the original 68 ones. And give them different weights to computer the final score to rank the schools.

The PCA (principle component analysis) is an ideal method to tackle this problem with high efficiency [2]. PCA uses linear equation to combine the original indicators, generating principle components F_i with the maximized variance. The variance of a component was defined as its amount of information.

5.2.1 Standardization

Firstly, we recognize that the original data are not complete. We use the mean value to fill up the data, where i is the number of the school, j is the

number of the indicator. This treatment is reasonable because it doesn't influence the result [3].

$$NULL_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij} = \overline{x}_j \quad (i = 1,2,\ldots,n; j = 1,2,\ldots,p) \tag{1}$$

Secondly, we use the equation below to standardize the data:

$$x_{ij}^{*} = \frac{x_{ij} - \overline{x}_j}{\sqrt{Var(x_j)}}$$
$$Var(x_j) = \frac{1}{n-1}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2 \tag{2}$$

5.2.2 Calculation

We use the equation below to compute the covariance matrix where i, j both are the indicators and x are the data after standardization.

$$r_{ij} = \mathrm{cov}(x_i, x_j) = \frac{\sum (x_i - \overline{x}_i)(x_j - \overline{x}_j)}{n-1} \tag{3}$$

And then we calculate the eigenvalue and eigenvector of the covariance matrix:

Eigenvalue: λ_i; Eigenvector: $\vec{a}_i$, $(i = 1,2,\ldots,p)$

5.2.3 Principle Components

The steps above can generate 68 principle components while we can only utilize 3 or 4 of the components, which have the highest contribution rate. The contribution rate can be calculated by the equation:

$$Contribution\ rate = \frac{\lambda_i}{\sum_{i=1}^{p}\lambda_i} \tag{4}$$

The sum of the contribution rate is denoted by T, the cumulative contribution rate. The higher *T*, the more information contained in the principle components.

At last, we can sum the scores of the principle components and use the total score to rank the schools.

5.2.4 PCA Results

We set the cumulative contribution rate $T = 90\%$, the 2323 schools and 68

indicators as inputs. And apply the PCA method to compute the score of the schools, the top 500 schools will go around to next part. Here is the table of the top ten schools.

Rank	*UNITID*	*INSTNM*	*PCA Score*
1	*244190*	*Widener University-Delaware Campus*	*32.87*
2	*227429*	*Paul Quinn College*	*30.15*
3	*138761*	*Andrew College*	*27.14*
4	*225575*	*Huston-Tillotson University*	*25.09*
5	*102270*	*Stillman College*	*24.99*
6	*140720*	*Paine College*	*21.81*
7	*447582*	*New River Community and Technical College*	*21.58*
8	*198862*	*Livingstone College*	*21.30*
9	*229063*	*Texas Southern University*	*20.43*
10	*233338*	*Richard Bland College of the College of William and Mary*	*20.21*

Table 3: top ten candidate schools by PCA

In the table above, the school that has the highest PCA scores has the largest demand of grant based on our focus. Please notice that this is not the final recommending list of the schools, because we haven't considered the ROI of these schools. After deriving the ROI function, we will offer a list based on the potential use of the investment. Further more, we will compare the difference between the PCA rank list between the final list later.

Checking the statistics of the top school *Widener University-Delaware Campus*, we find that the students in this school are under a large amount of debt. It is the main indicator pushing the school to the top one. To some degree, it reflects that our model is reasonable and efficient for we pay attention to those schools whose students are under much debt. On another hand, our PCA-ranking model has a clear, easy-to-understand basis in focus correlation measures and gives reasonably accurate results. It should be noted that we choose this approach to ranking schools over a much simpler approach such as simply summing the indicators for various reasons, one of which is that there are interactions between the indicators and PCA is skilled at solving this kind of problems.

6 Strategy Making

The model we created in the sections above works well for selecting the target schools. In this part, in order to rank the target schools and determine the investment strategy, we must determine the return on investment (ROI), before this, we should derive a utility function which measures the contribution of the school to the society. ROI is defined as the difference between the utility before and after the investment.

The utility of a school is an abstract value, and it has no real unit for it's not money, not earnings or other things. And the input *x* is a vector, including four main factors, student number, graduated ratio, earnings of graduated students and the investment. The construct of *x* will be explained later. Here is the key assumption: the utility U_i and the inputs *x* have the relationship below:

$$U_i = \log(x) \tag{5}$$

First of all, the utility function must be concave for two reasons:

1. If the function is linear or convex, the best strategy is to give the whole 100 million to one school to maximize the total utility.
2. The utility is a state value. If both a high state and a low state school receive a same amount of investment, the low state school will generate more incremental utility. Because there are more opportunities to use the money and more things to be improved in the low state one.

Secondly, motivated by the utility function in microeconomics, we use the "log" relationship to determine the utility function. Although the relationship may be more complex in real life, it gives us reasonable ROI function below and works well with our model.

6.1 The ROI Function

Here we come to the ROI function, the function describes the relationship between investment and incremental utility. Once the ROI function is decided, we can make the investment strategy to maximize the total ROI.

Naturally, the ROI function is the difference of the utility function:

$$ROI_i = \Delta U_i = \log(x_0 + \Delta x_i) - \log(x_0) \tag{6}$$

We can plot a figure to describe the relationship:

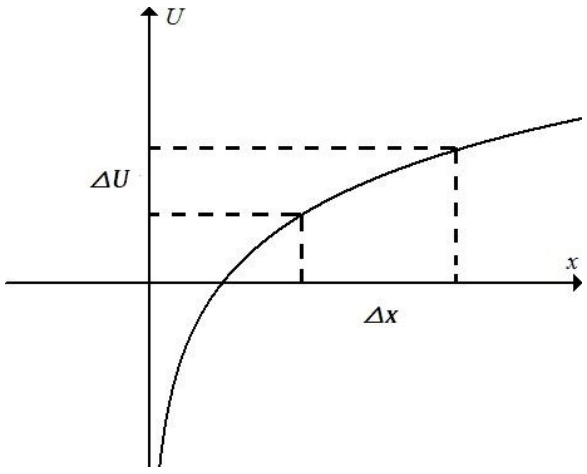


Figure 3: the relationship between incremental utility and investment

In Figure 3 and equation (6), Δx_i is the incremental amount of the inputs triggered by the investment. In the following steps, we will derive an explicit equation of *x*.

Firstly, we consider the number of the students *N*, median of earnings after 10 years *M* and the ratio of students who earn more than 25k per year after 6 years *m*. We multiply them together to measure the total earnings [4]: $m \cdot M \cdot N$. The earnings measure one aspect of the utility, the higher it is, the higher the state of school will be.

Secondly, we take a deep look at the graduation rate *g* (same for two year school or four year school). This indicator measures another aspect of the utility. It also has a positive relationship between the utility. We intend to combine the two aspects together but the units are not equal. Then, we define a constant *k* to balance the unit. As a result, the utility function can be written as the form below:

$$U_0 = \log(m \cdot M \cdot N + k \cdot g) \tag{7}$$

Here all the variables are different for each school, for simplicity, we eliminate the subscript *i*.

Take the investment into consideration, the utility will be:

$$U_j = \log(\sum_{p=0}^{j-1} x_p + x_j + m \cdot M \cdot N + k \cdot g) \tag{8}$$

where *j* represents the time period, $j = 1,2,3,4,5$ and $x_0 = 0$. x_0 is the initial investment the school received from other foundations or the government. In this problem, we simply assume $x_0 = 0$ because of the limited data.

At last, we can write down the ROI function:

$$ROI = \begin{cases} \log(x_j + m \cdot M \cdot N + k \cdot g) - \log(m \cdot M \cdot N + k \cdot g) \cdots j = 1 \\ \log(\sum_{p=0}^{j-1} x_p + x_j + m \cdot M \cdot N + k \cdot g) - \log(\sum_{p=0}^{j-2} x_p + x_{j-1} + m \cdot M \cdot N + k \cdot g) \cdots j = 2,3,4,5 \end{cases} \tag{9}$$

The ROI function above is dynamic with respect to time period. In other words, when time changes, the ROI function changes in response. To be more specific, once we invest in a school, the ROI in the next period will decrease. The larger amount of investment there is, the lower the ROI will be. This character of our ROI function describes the real situation precisely, which makes model close to the reality.

6.2 Optimizing the Total ROI

We now assign 100 million dollars to 50 target schools to maximize the total

ROI. Each school's ROI is defined in Equation (9). We use the equations below to simplify the form of our ROI function:

$$\begin{aligned} c_j &= \sum_{p=0}^{j-1} x_p + m \cdot M \cdot N + k \cdot g \cdots\cdots j \geq 1 \\ c_o &= m \cdot M \cdot N + k \cdot g \cdots\cdots\cdots\cdots\cdots j = 0 \end{aligned} \tag{10}$$

The utility function will be:

$$U_j = \begin{cases} \log(x_j + c_j) \cdots\cdots\cdots\cdots\cdots\cdots j \geq 1 \\ \log c_0 \cdots\cdots\cdots\cdots\cdots\cdots\cdots j = 0 \end{cases} \tag{11}$$

The ROI function will be:

$$ROI = \Delta U_j = U_j - U_{j-1} \tag{12}$$

We want to maximize the total ROI, the function can be:

$$\begin{aligned} &\sum_i \sum_j \Delta U_{ij} \\ &= \sum_i [(U_{i1} - U_{i0}) + (U_{i2} - U_{i1}) + (U_{i3} - U_{i2}) + (U_{i4} - U_{i3}) + (U_{i5} - U_{i4})] \\ &= \sum_i (U_{i5} - U_{i0}) \\ &= \sum_i U_{i5} - \sum_i U_{i0} \end{aligned} \tag{13}$$

The result is really exciting! It is noted that the ROI of one school in different time period cancels out with each other. As $\sum_i U_{i0}$ is constant, the problem will be in this form:

$$\begin{aligned} &\max \sum_i U_{i5} = \sum_i \log(x_{i1} + x_{i2} + x_{i3} + x_{i4} + x_{i5} + c_{i0}) \\ &s.t. \begin{cases} \sum_i x_{i1} = 1 \times 10^8 \\ \sum_i x_{i2} = 1 \times 10^8 \\ \sum_i x_{i3} = 1 \times 10^8 \\ \sum_i x_{i4} = 1 \times 10^8 \\ \sum_i x_{i5} = 1 \times 10^8 \\ x_{ij} \geq 0 \end{cases} i = 1,2,\cdots,50; j = 1,2,\cdots,5 \end{aligned} \tag{14}$$

The equations (14) are the final form of our model. Up to know, we have already derived a mathematical problem of the real situation. Now, we attempt to choose the best method to solve the optimization problem.

6.2.1 Karush-Kuhn-Tucker Conditions

At first, we set out to implement KKT conditions to solve the optimization problem [5]. For the problem below:

$$\begin{aligned} &\min f_0(x) \\ &f_i(x) \le 0, i = 1,2,\cdots\cdots m \\ &h_i(x) = 0, i = 1,2,\cdots\cdots p \end{aligned} \tag{15}$$

the KKT conditions are presented as follows:

$$\begin{cases} \nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) = 0 \\ f_i(x^*) \le 0 \\ h_i(x^*) = 0 \\ \lambda_i^* \ge 0 \\ \lambda_i^* f_i(x^*) = 0 \end{cases} \tag{16}$$

where λ_i is Lagrange multiplier associated with $f_i(x) \le 0$;

ν_i is Lagrange multiplier associated with $h_i(x) = 0$.

For our problem, we can easily transform the formulation into the standard form, and the corresponding KKT conditions are:

$$\begin{cases} \forall i, \dfrac{1}{x_{i1}^* + x_{i2}^* + x_{i3}^* + x_{i4}^* + x_{i5}^*} - \sum_{i=1}^{m} \lambda_i^* + \sum_{i=1}^{p} \nu_i^* = 0 \\ x_{ij} \ge 0 \\ \sum_{ij} x_{ij} = 1 \times 10^8, j = 1,2,3,4,5 \\ \lambda_i^* \ge 0 \\ \lambda_i^* f_i(x^*) = 0 \end{cases} \tag{17}$$

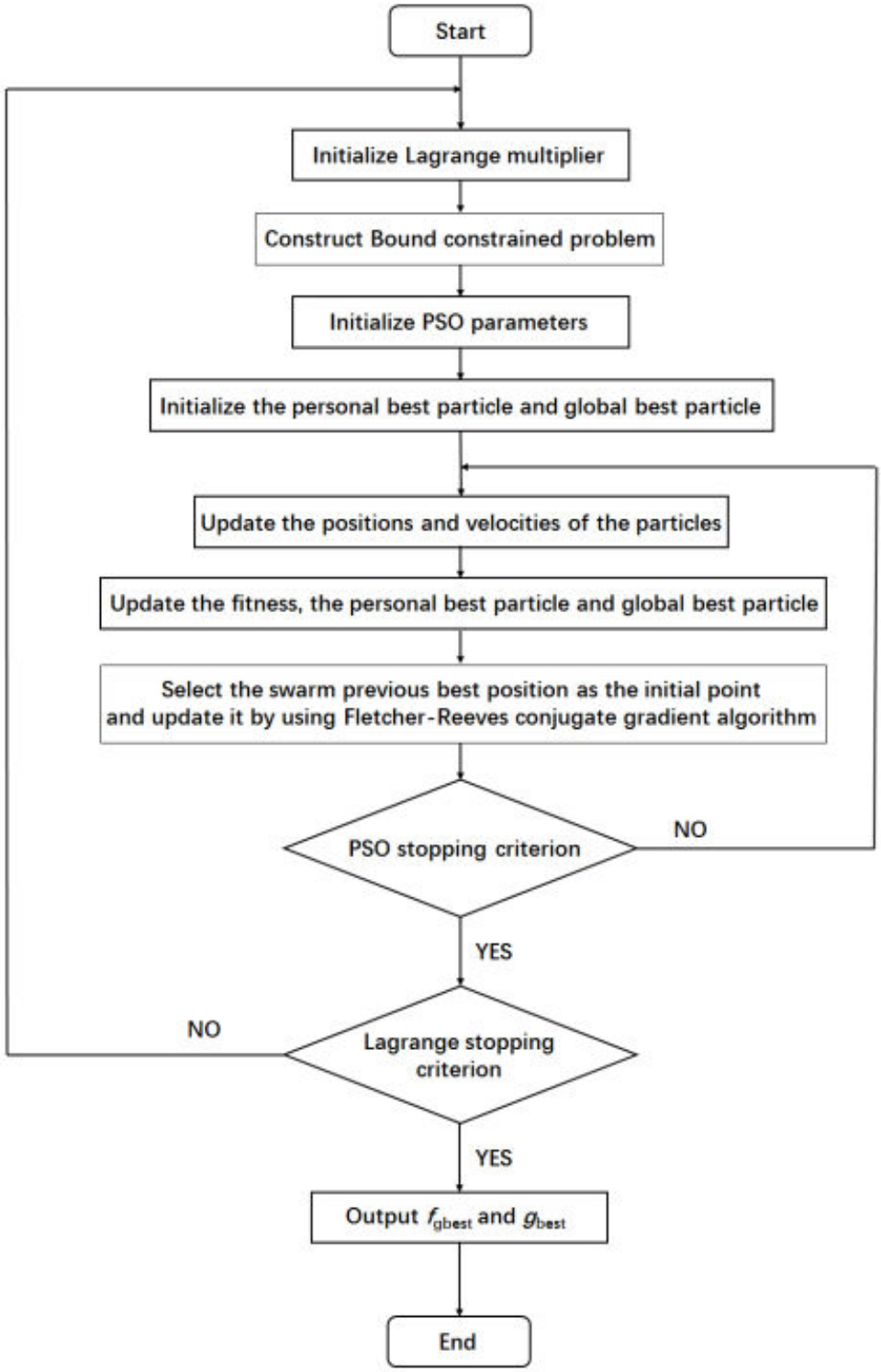


Figure 4: flow chart of the LA_PSA_GT algorithm

f_{gbest} is the maximum of the total ROI function

g_{best} is vector of investment amount per year

The KKT conditions ensures the existence of the solutions. Solving the equations above, we will absolutely get the optimized strategy. However, the equations are extremely difficult since there are 250 variables. In general, the KKT method is highly efficient dealing with small number of variables. So, as a result, the number of the variable is too big for us to solve the equation when it comes to our problem. In other words, we are not able to create a program to solve the equations. Consequently, we have to give up and search for other algorithms.

6.2.2 PSO Algorithm

The PSO (particle swarm optimization) is an efficient method, imitating the flying path of a bird swarm. PSO algorithm uses three flying principles, collision avoidance, velocity matching and flock centering to search the extrema within the boundary conditions. The algorithm starts with a randomly initialized investment vectors and iterate them until reaching the maximum point [6]. We create a program based on the PSO, but the algorithm takes a very long time to converge and does not produce the optimal values. Therefore, we decide to use another method to improve the performance.

6.2.3 Improved PSO algorithm based on augmented Lagrange function (LA_PSO_GT)

We search for other literary researches and decide to use the method above. The algorithms utilize the augmented Lagrange function to change the optimization problem into boundary constraint optimization problem. After that, the LA_PSO_GT combines Conjugate Gradient method and PSO algorithm together, which overcome the defect of the length of the convergence time and improves the computing efficiency [7]. The whole procedures will be attached in the appendix. Above is the flow chart of this algorithm.

7 Result

After running the program, we get the amount of investment for each target school in each year and the ROI in five years.

The fitness figure of the algorithm is presented below:

From the fitness figure below, we can see that the algorithm converges after 50 iterations. It proves that our method is efficient and powerful.

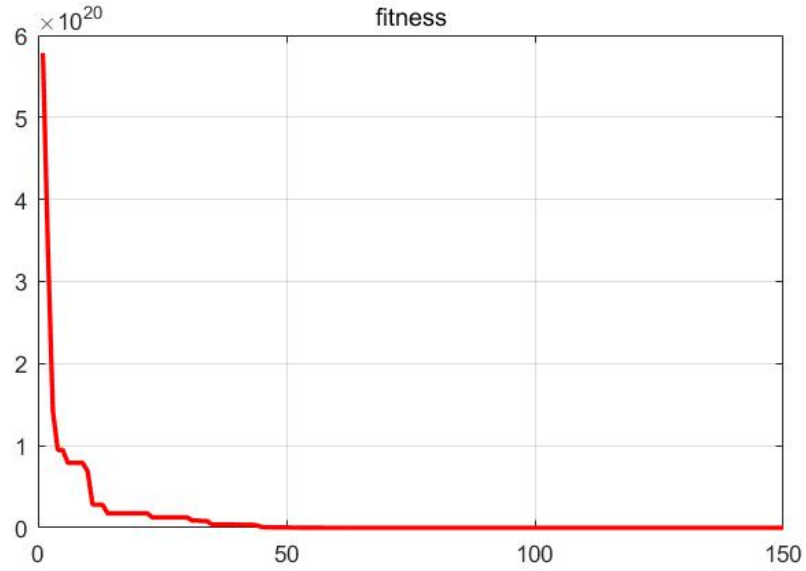


Figure 5: fitness of the algorithm

For some schools, the investment is too small. For example, if only 1000 dollars is invested in a school in a year, it is obviously useless in real situation. Therefore, we eliminate the value and set it as zero. Although we don't use the whole 100 million, but the utilization rates are all above 99%.

7.1 Optimal Investment Strategy and Recommending List

Here is a table of top ten schools in the recommending list ranked by total ROI in five years, including the investment amount each year and the time duration. The full table of all target schools will be attached in appendix.

Rank	*INSTNM*	*ROI*	*t*	*1*	*2*	*3*	*4*	*5*
1	*MacCormac College*	*2.22*	*4*	*13.62*	*0.17*	*0.00*	*5.69*	*4.78*
2	*Paul Quinn College*	*2.02*	*5*	*7.84*	*1.02*	*3.09*	*2.92*	*5.10*
3	*Andrew College*	*1.81*	*3*	*0.00*	*0.00*	*0.33*	*5.11*	*7.94*
4	*Bidwell Training Center Inc*	*1.80*	*4*	*0.18*	*0.00*	*7.18*	*0.66*	*0.75*
5	*Colorado Heights University*	*1.64*	*3*	*8.24*	*0.46*	*11.32*	*0.00*	*0.00*
6	*Tougaloo College*	*1.12*	*5*	*12.01*	*0.31*	*0.44*	*5.62*	*2.29*
7	*Brevard College*	*1.05*	*4*	*5.46*	*0.00*	*9.13*	*0.22*	*1.85*
8	*Paine College*	*0.98*	*4*	*0.00*	*0.98*	*3.48*	*6.12*	*0.43*
9	*Fisk University*	*0.90*	*5*	*8.84*	*1.49*	*0.70*	*0.71*	*0.14*
10	*Southeast Missouri Hospital College of Nursing and Health Sciences*	*0.84*	*4*	*2.38*	*1.32*	*0.00*	*5.59*	*3.80*
...	*...*	*...*	*...*	*...*	*...*	*...*	*...*	*...*
	Utilization rate of funds in each year(%)			99.60	99.79	99.63	99.71	99.69

Table 4: the top ten schools and their investment amount in our recommending list (unit:million)[8]

As we can see in the recommending list, two of the schools, *Paul Quinn*

College and *Andrew College* are also the top ten schools in the PCA result. Note that we use different data to rank the schools, it proves that the data has correlation. What's more, it shows that two results are consistent with each other. The target schools are those with high ROI and worth investmemt.

8 Testing our Model

To test our model whether it duplicates the investment and focus of Gates Foundation or not, we collect data of the investment distribution of the two foundations, and compare them to our recommending list.

Rank	*INSTNM*	*Investment amount (million)*
1	*Harvard University*	*24.14*
2	*University of Washington*	*16.42*
3	*Seattle University*	*15.80*
4	*Yale University*	*7.50*
5	*Texas Tech University*	*6.96*
6	*University of Michigan*	*6.85*
7	*Land-Grant Universities*	*4.95*
8	*University of Kentucky*	*4.50*
9	*Columbia University*	*3.78*
10	*Stanford University*	*3.25*

Table 5: the top ten schools ranked by Gates Foundation's investment amount

In the table above, we can see that Gates Foundation invested mostly on those well-known schools like *Harvard* and *Yale*. But none of these schools appear in our recommending list. It proves that our model efficiently eliminate other foundations' investment and focus.

8.1 Sensitivity Analysis

In the process of determining the ROI function, we create a constant k to make the unit equal while this parameter lacks database. We use the ratio of mean earnings and mean graduation rate to determine the value of k. To some degree, k is the marginal rate of substitution between the two factors. How does the change of k influence our outputs? We analyze the standard deviation of the invest mount caused by a slightly change of k.

Year	$\boldsymbol{k-10\%}$	$\boldsymbol{k-5\%}$	$\boldsymbol{k+5\%}$	$\boldsymbol{k+10\%}$
1	*12.18%*	*5.33%*	*4.52%*	*9.51%*
2	*9.09%*	*4.25%*	*5.64%*	*11.88%*
3	*10.88%*	*6.37%*	*5.63%*	*13.28%*
4	*7.64%*	*3.29%*	*3.27%*	*13.33%*

5	*12.67%*	*4.90%*	*4.94%*	*7.41%*
Mean value	*10.49%*	*4.83%*	*4.80%*	*11.08%*

Table 6: the standard deviation of the investment amount correspond to the change of MRS *k*

From the table above, we can see that the influence of *k* is not very large, and within the range we can bear.

8.2 Strengths

1. **Data classification**. When faced with big data problems, the pattern classification is extremely useful and important. Based on this insight, our model firstly implements a data classification, separating the data for two major purposes, one for selection, another for determining the ROI function. By this means, we make the most use of the data attached.
2. **Data extraction**. Our model utilizes the PCA method to select the target schools based on the focus of our foundation. The data selection is also an important part in big data problem, which will improve the efficiency for further data processing.
3. **Reasonable ROI function**. Another strength of our model is that we create an ROI function by analyzing the relationship between earnings and graduation rate. Furthermore, our function can easily be modified if other researches find the accurate relationship.
4. **Suitable algorithm**. In the third part, we derive a mathematical problem to describe the strategy in a general form. In order to solve the math problem, we have tried several algorithms and finally decide to choose the most suitable one.

8.3 Weaknesses

1. **Data vacancy**. There are too many vacancies (NULL) in the data attached. And we fill in the blanks with the mean value of this column, which is a main source of errors.
2. **Parameter without database**. When determining the ROI function, we create a constant *k* to balance the unit, where *k* is the marginal rate of substitution. The value *k* will influence our outputs to some degree. So we made a sensitivity analysis to study the influence of *k*.
3. **Algorithm limitations**. Although the PSO algorithm is a suitable method to compute our model, it has several limitations. The local search capacity of PSO is not very well, so the accuracy of outputs is limited. On the other hand, the algorithms may converge too early before the particles find the global optimization points. Although the particles are

near the final positions, they lack the capacity to jump out of the situation because of the low velocity.

9 Conclusion

We have been asked by the Goodgrant Foundation to determine the optimal investment strategy base on the potential and ROI of the candidate schools. Since it is a big data problem, we use data classification, data selection as pretreatment, after which we decide the investment focus of the foundation and the ROI of the schools. Finally, we make the optimal strategy by PSO algorithm.

We deeply analyze the types, meanings, and interactions of the data. Then, by deciding our focus for charitable meanings, we attempt to invest much on those schools with more minority races, low SAT scores, high debt ratio and so on. We want to help those schools who don't performance well as they have the highest likelihood of producing a strong positive effect on student performance.

Then, we select the target schools by PCA based on the key indicators we focused. PCA ranks the candidate schools by the correlation with our concern, and we select the top 50 of them as target schools. After that, we determine an estimated ROI function to quantify the incremental utility of the schools that receive our investment based on the earnings and graduate rate.

Using an iterative, multivariable, machine-learning algorithm, we are able to optimize the total ROI of our target schools in five years. The solutions combine the investment amount and time duration of each school. Since we have the investment amount already, we can calculate the cumulative ROI in five years of each school and offer a recommending list. The top schools in the list are not well known schools, but the schools in village, small cities and so on, which correspond to our foundation's focus.

10 Letter to the CFO of the Goodgrant Foundation

Dear Mr. Alpha Chiang,

In response to your questions regarding the optimal investment strategy to improve the performance of undergraduate students, we are writing to inform you of our work.

We first decide the focus of your foundation. Since your foundation is a charitable organization intending to improve the educational performance, we recommend you to invest on the schools whose students has lower SAT/ACT scores, under much debt, from minority races and so on. This concern reflects your charitable identity and has the higher potential to improve the performance.

Then, we rank the schools by the correlation with the focus. If a school has these characters above, it will come first.

The most creative work is that we determine a reasonable ROI function based on the social utility. We use "utility" to measure the contribution of a school. Firstly, we notice that if a school have more students, the contribution of the school is higher (assume other conditions are same). Because the school trains more students to workers. Secondly, we notice that the higher the earnings of graduated students are, the higher the utility is. So we multiply the number of students and the median of earning together to measure the total earning. Thirdly, we also notice that a higher utility school has high graduation rate. Therefore, we add the earnings and the graduation rate together as the total inputs. The investment amount will be added to the part of the earnings. Further more, we create a marginal rate of substitution to measure the interaction of the two factors. We can't find the accurate value of the MRS but can be modified by deep researches.

The ROI function is the incremental utility after the investment, which is easy to understand. And another point is that the utility function is concave, which means the ROI will be less if invest more on a school. In other words, if we invest the same amount on a certain school, the ROI will be less than this year. So the schools with less earnings and lower graduation rate have large ROI, which means the potential of the effective use of funding is high. We think that this model describe the real situation well.

Another accomplishment is that we utilize a complex algorithm to maximize the total ROI of the target schools. The algorithm is typically useful to solve the problem to determine the future investment amount and time duration for each target school. Further more, our algorithm can easily be changed to solve other investment strategy making problems with different amount of donations and different ROI functions. In other word, we offer a general method to solve this kind of problems.

Almost none of our invested schools are well-known, which is consistent with our focus, because the well-known schools receive so much investment from other organizations like Gates Foundation. The complete investment strategy is attached in the appendix.

To summarize, we cling to the belief that our model is a powerful tool to help you devise the best investment strategy, including the amount of investment and time duration per school.

Yours, sincerely

Team 47823

11 References

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[8]Retrieved January 31, 2016, from http://www.gatesfoundation.org/How-We-Work/Quick-Links/Grants-Database#

12 Appendix 1: Recommending List

Rank	INSTNM	ROI	t	1	2	3	4	5
1	MacCormac College	2.22	4	13.62	0.17	0.00	5.69	4.78
2	Paul Quinn College	2.02	5	7.84	1.02	3.09	2.92	5.10
3	Andrew College	1.81	3	0.00	0.00	0.33	5.11	7.94
4	Bidwell Training Center Inc	1.80	5	0.18	0.00	7.18	0.66	0.75
5	Colorado Heights University	1.64	3	8.24	0.46	11.32	0.00	0.00
6	Tougaloo College	1.12	6	12.01	0.31	0.44	5.62	2.29
7	Brevard College	1.05	5	5.46	0.00	9.13	0.22	1.85
8	Paine College	0.98	5	0.00	0.98	3.48	6.12	0.43
9	Fisk University	0.90	6	8.84	1.49	0.70	0.71	0.14
10	Southeast Missouri Hospital College of Nursing and Health Sciences	0.84	5	2.38	1.32	0.00	5.59	3.80
11	Stillman College	0.83	3	0.00	1.95	2.75	6.29	0.00
12	Livingstone College	0.75	3	0.00	6.15	0.00	0.78	3.92
13	Leech Lake Tribal College	0.73	2	2.00	3.60	0.00	0.00	0.00
14	Virginia Union University	0.70	5	1.95	10.62	2.42	1.70	0.32
15	Widener University-Delaware Campus	0.69	3	0.00	2.02	0.11	1.98	0.00
16	University of South Carolina-Salkehatchie	0.66	5	2.10	2.58	0.25	0.30	1.28
17	Elizabeth City State University	0.66	5	3.26	8.59	6.22	2.30	4.88
18	Huston-Tillotson University	0.65	4	1.32	7.13	0.39	0.00	3.80
19	Edward Waters College	0.64	4	0.00	3.49	1.08	1.89	0.24
20	Claflin University	0.62	4	2.46	4.10	5.49	0.00	5.61
21	Sterling College	0.60	3	0.00	0.00	4.92	0.26	3.06
22	Olivet College	0.60	4	0.00	3.49	5.29	2.33	3.47
23	Ohio Valley University	0.51	2	1.35	0.00	3.40	0.00	0.00
24	Rust College	0.49	5	0.30	0.46	1.10	0.10	1.67
25	Culver-Stockton College	0.48	4	0.00	9.01	0.16	1.78	0.26
26	Bethany College	0.47	4	0.00	1.55	0.30	6.24	0.82
27	Greensboro College	0.46	4	0.00	0.43	5.36	0.11	4.16
28	Ottawa University-Ottawa	0.46	3	0.00	0.38	0.00	8.62	0.20
29	Webber International University	0.44	3	0.00	6.00	1.43	0.17	0.00
30	Bacone College	0.42	3	2.72	0.00	0.45	4.32	0.00
31	Dillard University	0.38	5	0.53	1.10	1.98	3.74	1.86
32	North Greenville University	0.38	5	2.10	0.20	3.70	0.26	9.26
33	Manor College	0.36	4	0.00	2.31	0.12	5.08	0.66
34	Newberry College	0.36	5	0.35	0.44	0.70	0.80	5.42
35	New River Community and Technical College	0.36	5	0.15	3.26	0.49	1.26	8.15

36	*Orleans Technical Institute*	*0.31*	*4*	*0.89*	*1.19*	*0.00*	*0.16*	*0.36*
37	*Louisburg College*	*0.27*	*3*	*0.00*	*0.81*	*1.07*	*0.00*	*0.18*
38	*Bethune-Cookman University*	*0.21*	*3*	*8.43*	*1.15*	*0.00*	*0.78*	*0.00*
39	*Richard Bland College of the College of William and Mary*	*0.21*	*4*	*1.32*	*0.00*	*1.56*	*1.26*	*0.40*
40	*Chowan University*	*0.20*	*3*	*0.42*	*1.55*	*0.00*	*3.32*	*0.00*
41	*Alabama State University*	*0.16*	*5*	*1.69*	*0.00*	*0.54*	*5.06*	*1.28*
42	*Campbell University*	*0.12*	*5*	*2.59*	*0.61*	*6.94*	*0.10*	*5.85*
43	*Delaware State University*	*0.11*	*3*	*0.00*	*3.41*	*0.00*	*0.53*	*3.69*
44	*Texas Southern University*	*0.10*	*4*	*1.13*	*1.95*	*4.85*	*0.00*	*1.20*
45	*Cumberland University*	*0.09*	*3*	*0.37*	*0.00*	*0.11*	*2.02*	*0.00*
46	*Cochise College*	*0.09*	*2*	*2.61*	*2.36*	*0.00*	*0.00*	*0.00*
47	*Central State University*	*0.07*	*4*	*0.17*	*0.14*	*0.12*	*0.76*	*0.00*
48	*Alcorn State University*	*0.07*	*3*	*0.81*	*0.00*	*0.00*	*2.19*	*0.61*
49	*Saint Augustine's University*	*0.06*	*2*	*0.00*	*0.18*	*0.00*	*0.56*	*0.00*
50	*Chattanooga State Community College*	*0.02*	*2*	*0.00*	*1.83*	*0.65*	*0.00*	*0.00*
	Utilization rate of funds of each year (%)			*99.60*	*99.79*	*99.63*	*99.71*	*99.69*

13 Appendix 2: An Introduction to the Improved PSO Algorithm Based on Augmented Lagrange Function

(source: Li Desheng. Improvement and Application of Particle Swarm Optimization Coupling with Classic Optimization[D].Beijing: Beijing University of Civil Engineering and Architecture, 2014: 29-43.)

The algorithm attempts to solve the non-linear constraint optimization problem below:

$$\min: f(x)$$
$$s.t.\begin{cases} c_i(x)=0, i=1,2,\cdots,m_e \\ c_j(x)\geq 0, j=m_{e+1},\cdots,m \\ l\leq x\leq u \end{cases} \tag{1}$$

For this problem, if there is not the constraint $l\leq x\leq u$, we can use the augmented Lagrange multiplier method to solve it. Given the Lagrange multiplier vector λ^k and the barrier parameter vector σ^k, the sub-problem of the step k is:

$$\min P(x,\lambda^k,\sigma^k) \tag{2}$$

where:

$$P(x,\lambda,\sigma)=f(x)-\sum_{i=1}^{m_e}[\lambda_i c_i(x)-\frac{1}{2}\sigma_i(c_i(x))^2]-\sum_{i=m_{e+1}}^{m}\overline{P}_i(x,\lambda,\sigma) \tag{3}$$

$$\overline{P}_i(x,\lambda,\sigma)=\begin{cases} \lambda_i c_i(x)-\frac{1}{2}\sigma_i(c_i(x))^2, if\ \lambda_i-\sigma_i c_i(x)>0 \\ \frac{1}{2}\lambda_i^{\ 2}/\sigma_i, otherwise \end{cases} \tag{4}$$

If there is the boundary constraint $l\leq x\leq u$ in the non-linear constraint optimization problem, we modify the augmented Lagrange multiplier method above. We assume that we knew the Lagrange multiplier vector λ^k and the barrier parameter vector σ^k. So we solve the sub-problem in step k:

$$\begin{cases} \min P(x,\lambda^k,\sigma^k) \\ s.t. l\leq x\leq \mu \end{cases} \tag{5}$$

The Lagrange multiplier method is accomplished by inside and outside

to-tier construct. We solve the equations (5) inside the construct, generating a group of new initial variable. And modify the outside barrier parameter vector and the multiplier vector. Test whether satisfies the iteration criterion or not. If not, construct the sub-problem in the next iteration. If so, stop the algorithm.

When initialing the multiplier vector λ^0 and the barrier parameter vector σ^0, they are usually set as positive vector. In the process of iteration, we use the equations below to modify the Lagrange multiplier vector λ :

$$\begin{cases} \lambda_i^{k+1} = \lambda_i^k - \sigma_i^k c_i(\overline{x}^k), i = 1,2,\cdots m_e \\ \lambda_i^{k+1} = \max\{\lambda_i^k - \sigma_i^k c_i(\overline{x}^k), 0\}, i = m_{e+1},\cdots,m \end{cases} \tag{6}$$

where $\overline{x}^k$ is the solution of NO.k sub-problem (5), and we use the equation below to modify the barrier parameter vector:

$$\sigma^{k+1} = \gamma\sigma^k \tag{7}$$

When $\| \overline{c}(x^{k+1}) \|_2 \leq \frac{1}{4} \| \overline{c}(x^k) \|_2$, we set $\gamma = 1$; if not, we set $\gamma \succ 1$ (usually 10 or 100), where:

$$\| \overline{c}(x) \|_2 = \sqrt{\sum_{i=1}^{m_e} (c_i(x))^2 + \sum_{i=m_{e+1}}^{m} (\min\{c_i(x), 0\})^2} \tag{8}$$

Given an ε, the stopping criterion is $\| \overline{c}(x) \|_2 \leq \varepsilon$, stop the iteration of Lagrange multiplier method. $\overline{x}^k$ is an approximate optimization of (1)

The specific process of the improved PSO algorithm based on augmented Lagrange function:

Step A: initialize λ^0 and σ^0 ;

Step B: construct boundary constraint optimization problem (5);

Step C: initialize PSO parameters, including initial position and velocity;

Step D: compute the position $x_j(t+1)$ and velocity $v_j(t+1)$ of the next period;

Step E: update the personal best particle p_j and global best particle p_g;

Step F: set $x_0 = p_g, \gamma = 0.001, eps = 1.0e-3, d_0 = -g_0, kk = 0$

Step G: if $\| d_{kk} \| \leq eps$ or $kk > 2$, switch to Step L, or switch to Step H

Step H: set $\alpha = 1$;

Step I: $x_{kk+1} = x_{kk} + \alpha d_{kk}$

Step J: if $f(x_{kk+1}) < f(x_{kk}) - \gamma \alpha d_0 d_0^T$, switch to Step K, or set $\alpha = 0.5 * \alpha$, back to Step I;

Step K: compute $\beta_{kk} = \frac{g_{kk+1}^T g_{kk+1}}{g_{kk}^T g_{kk}}, d_{kk+1} = -g_{kk+1} + \beta_{kk} d_{kk}$, set $kk = kk + 1$, back to Step G;

Step L: update $p_g = x_{kk}$

Step M: judge the PSO stopping criterion, if so, switch to Step N, or back to Step D;

Step N: judge the Lagrange multiplier stopping criterion, if so, stop the algorithm, output the result, or switch to Step O;

Step O: modify the Lagrange multiplier parameter λ^k, σ^k according to (6)~(9)